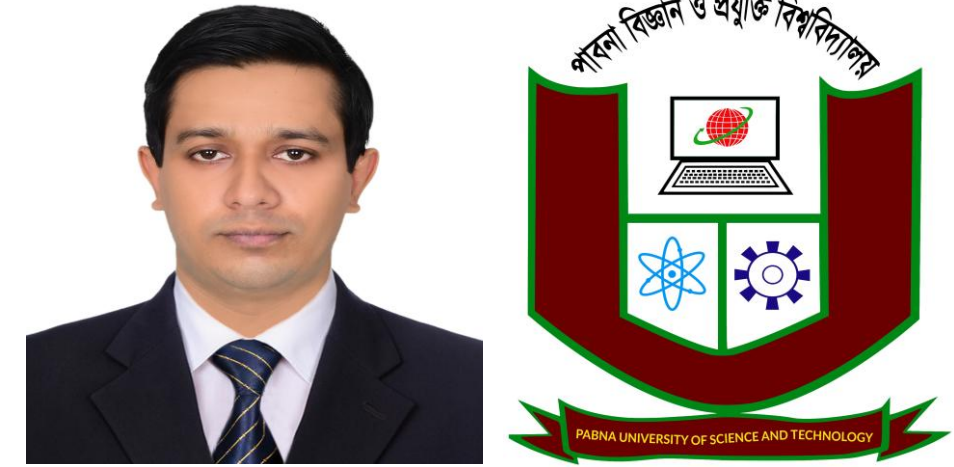




ABSTRACT

- ❑ We target automated five-class diabetic retinopathy grading to support early detection in low-resource settings like Bangladesh and contribute to SDG-3.
- ❑ We build a blended retinal fundus dataset from APTOS 2019 and IDRiD (6,740 images: 5,514 train, 592 validation, 634 test) with careful splits and augmentation.
- ❑ Images are resized, contrast-enhanced, circularly cropped, and color-normalized before extracting deep features and handcrafted descriptors.
- ❑ We compare frozen deep features, handcrafted-only, and early fusion strategies, fine-tune the CNNs, and train SVM, XGBoost, and ExtraTrees classifiers using multiple performance metrics.
- ❑ The best model (EfficientNet-B3 + SVM) achieves 72.7% accuracy, 58.4% macro F1, AUC 0.90, and QWK 0.59, indicating potential for scalable DR screening in resource-limited healthcare.

INTRODUCTION

Diabetic retinopathy (DR) is a major cause of preventable blindness, particularly in low-resource settings like Bangladesh where specialist screening is limited. It damages the retinal blood vessels and can lead to microaneurysms, hemorrhages, exudates, and neovascularization, eventually threatening central vision. Early detection traditionally depends on manual grading of retinal fundus images, which is time-consuming, subjective, and difficult to scale for large populations. Automated DR grading from fundus photographs can support frontline screening and contribute to progress toward SDG-3 on ensuring healthy lives and well-being.

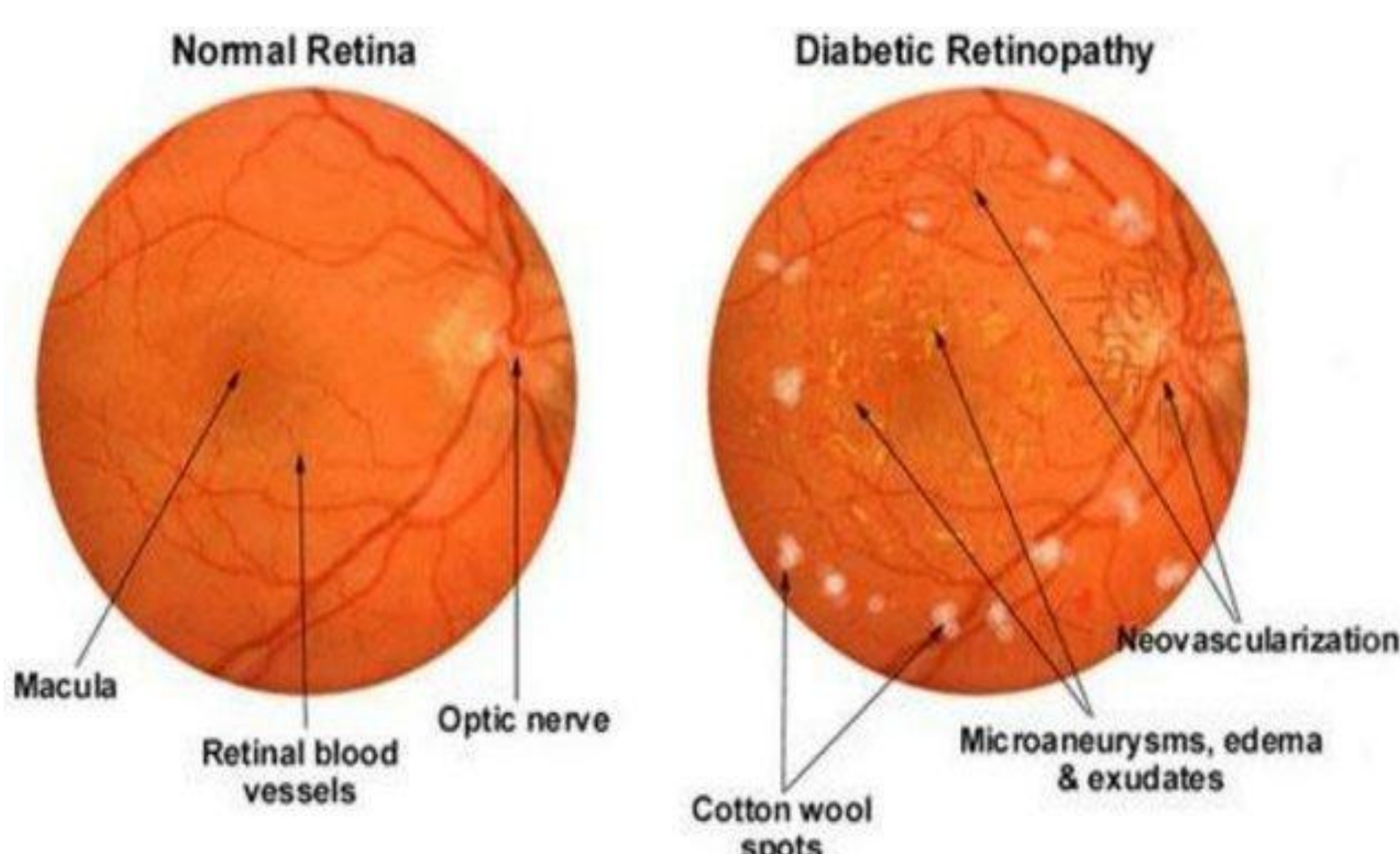


Figure 1: Normal retina versus diabetic retinopathy with microaneurysms, cotton wool spots, exudates and neovascularization.

OBJECTIVES

- Develop an automated five-class diabetic retinopathy grading framework using a blended APTOS 2019+IDRiD fundus image dataset.
- Compare deep CNN features, handcrafted descriptors, and their early fusion after standardized preprocessing and augmentation.
- Fine-tune and evaluate multiple model configurations using accuracy, precision, recall, macro F1, AUC, and quadratic weighted kappa with 95% confidence intervals.
- Assess the potential of the framework as a scalable DR screening component for resource-limited healthcare systems aligned with SDG-3.

DATASET

We use APTOS 2019 and IDRiD fundus images for five-class DR grading. After duplicate removal and augmentation on training images, blended APTOS–IDRiD splits yield 6,740 images (5,514 train, 592 validation, 634 test).

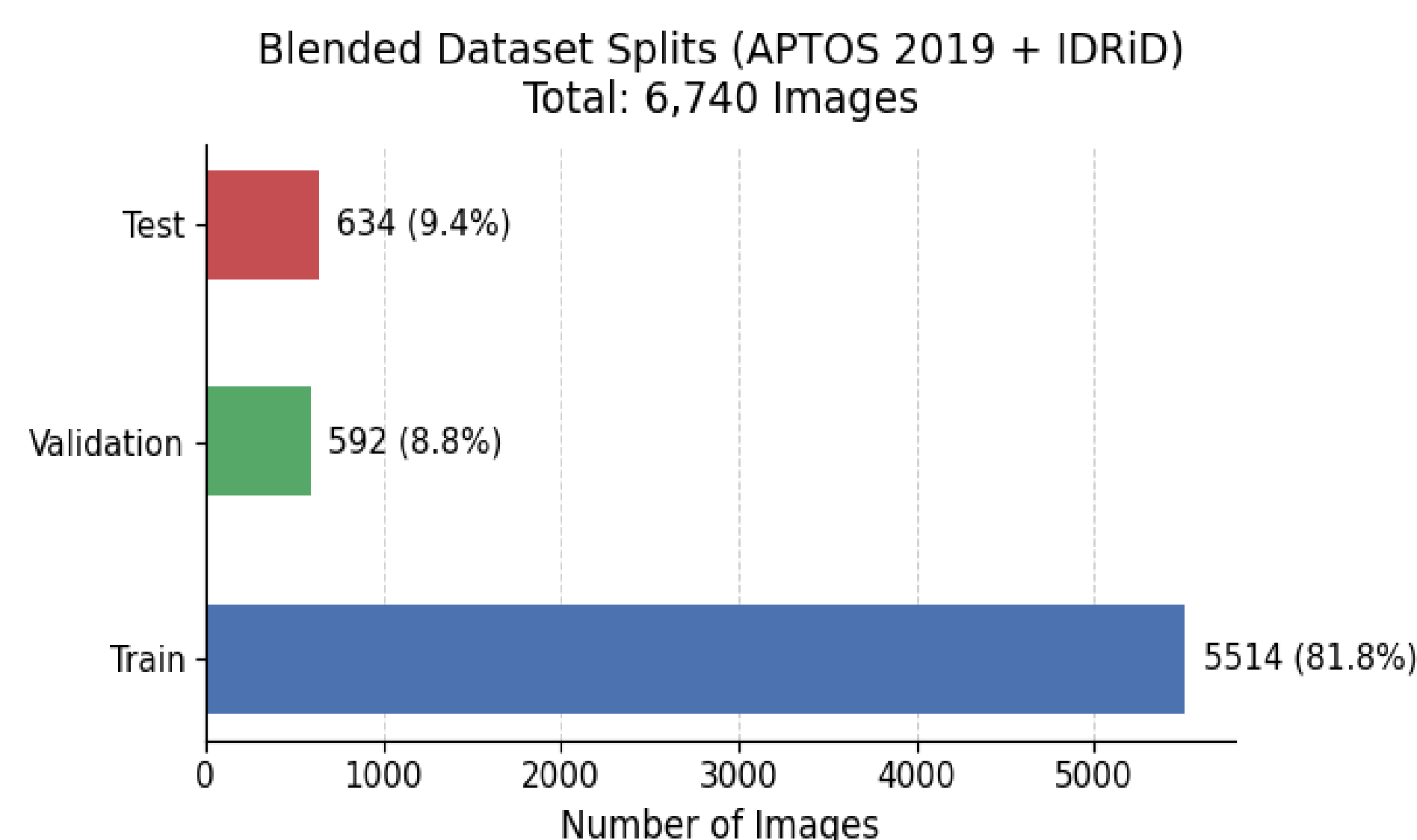


Figure 2: Final blended APTOS 2019 + IDRiD splits (train, validation, test; total 6,740 images).

DATA PREPROCESSING

We apply a standardized preprocessing pipeline. The main steps are:

- ❑ Deduplication of images
- ❑ Resizing to a fixed (224 x 224) input resolution
- ❑ Denoising and Gaussian blur to suppress sensor noise
- ❑ Contrast enhancement with CLAHE
- ❑ Circular cropping to isolate the retinal region
- ❑ Color normalization across images
- ❑ Data augmentation (e.g., flips, rotations, intensity changes) applied only to training images

PROPOSED FRAMEWORK



Figure 3: Proposed methodology framework.

EXPERIMENTAL SETUP

- ❖ Implemented in Python on a GPU-enabled workstation with mini-batch training and early stopping on validation performance.
- ❖ All models trained on the same blended train/validation splits and evaluated once on the held-out blended test set.
- ❖ Each configuration run multiple times; results reported as mean $\pm$ SD with 95% CIs.
- ❖ Metrics: accuracy, macro precision/recall, macro F1, AUC, and quadratic weighted kappa (QWK).

RESULTS

Macro F1 on Blended Test Set (Main SVM Configurations)

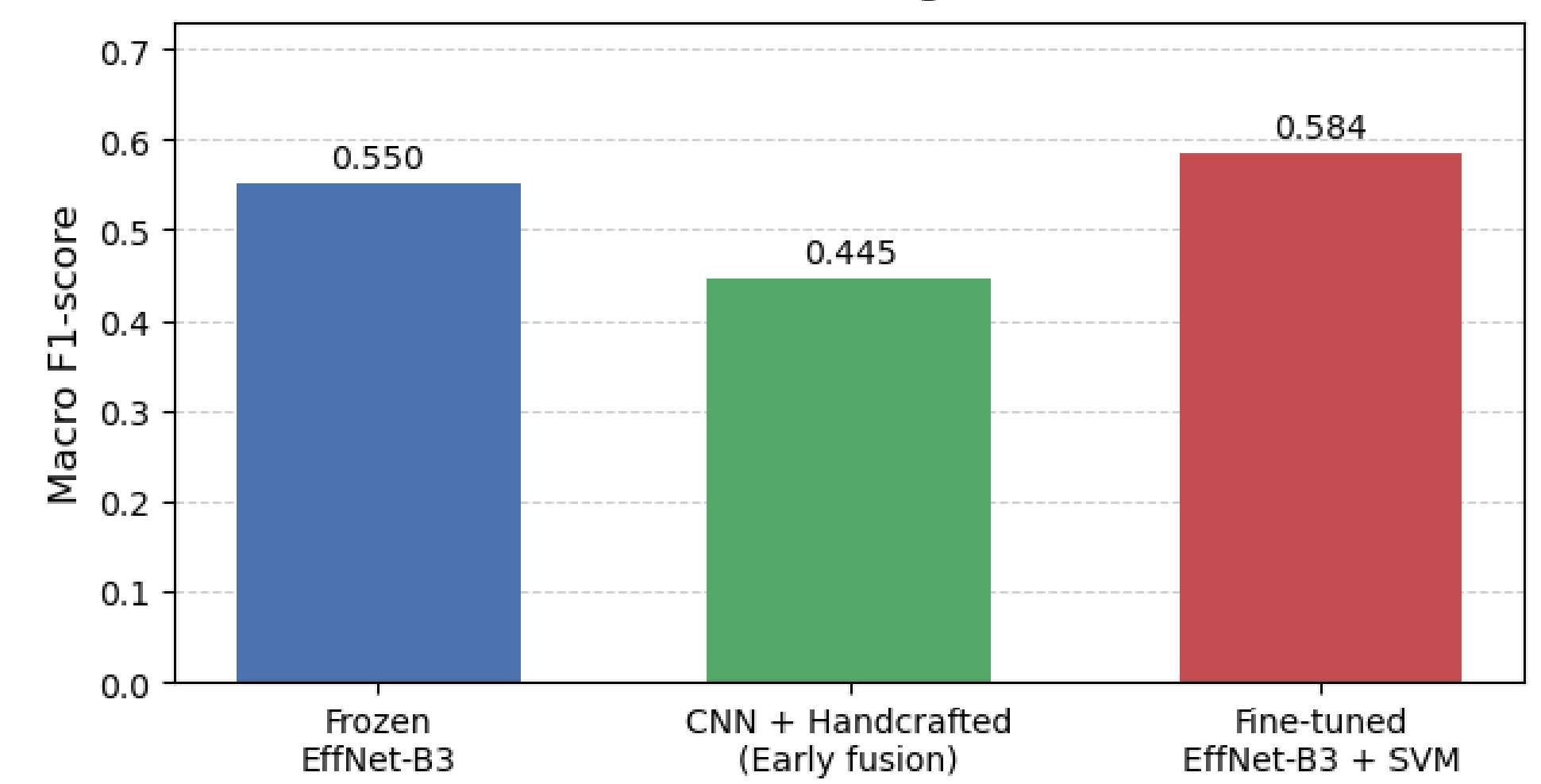


Figure 4: Macro F1 of main SVM configurations.

Agreement (QWK) for Fine-tuned EffNet-B3 with Different Classifiers

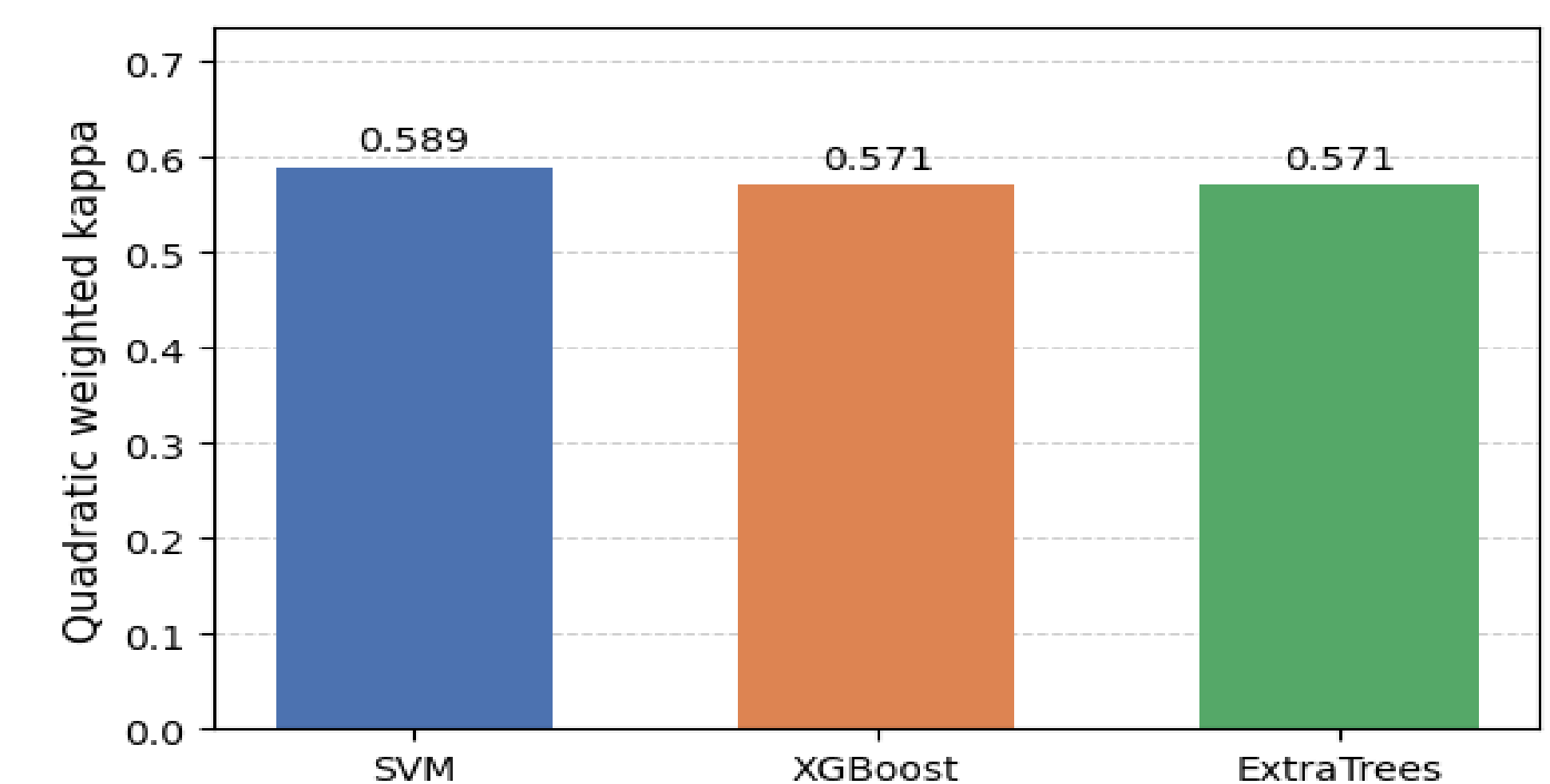


Figure 5: QWK for classifiers on fine-tuned EfficientNet-B3.

Accuracy on Blended Test Set (Main SVM Configurations)

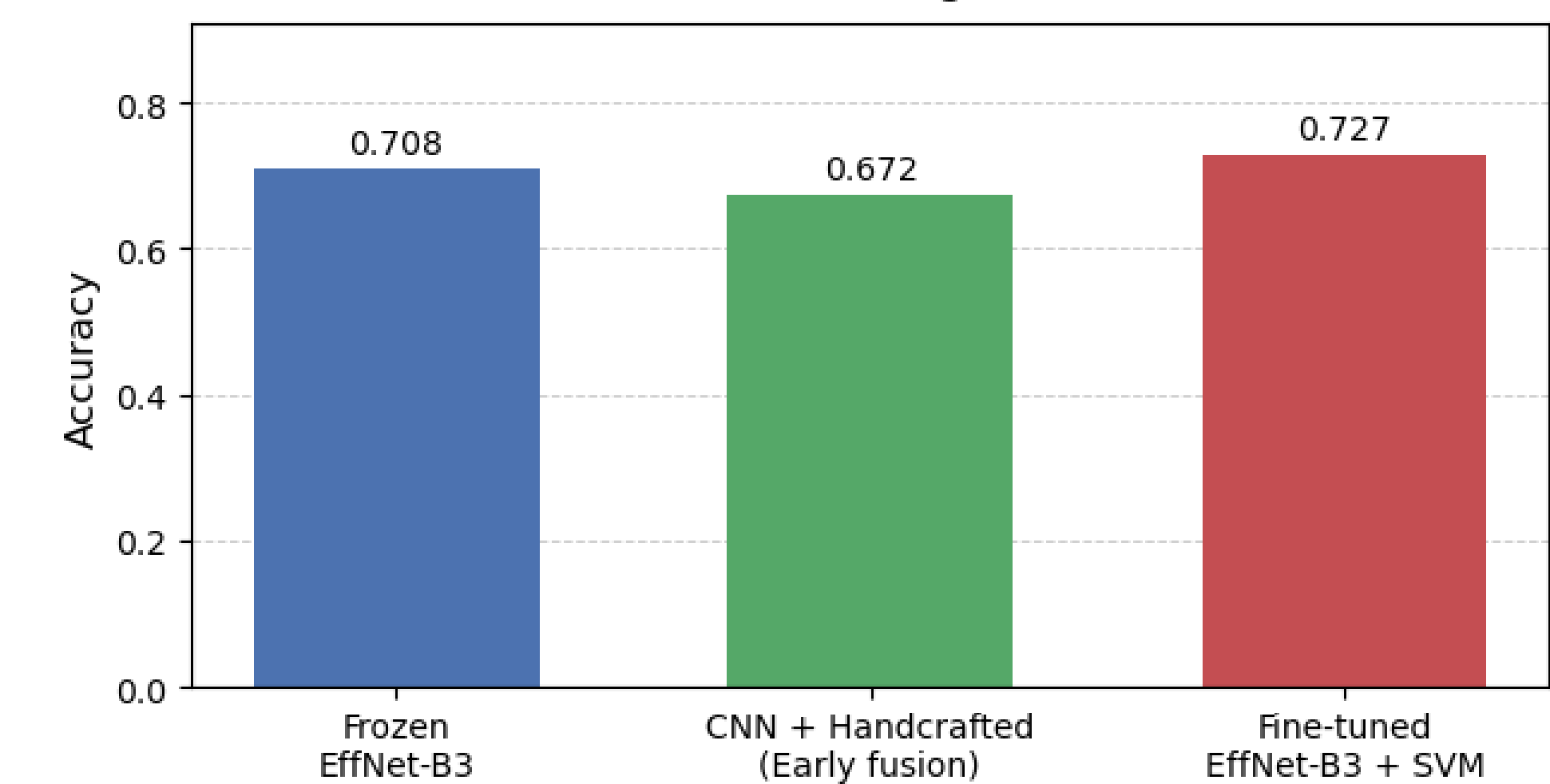


Figure 6: Accuracy of main SVM configurations.

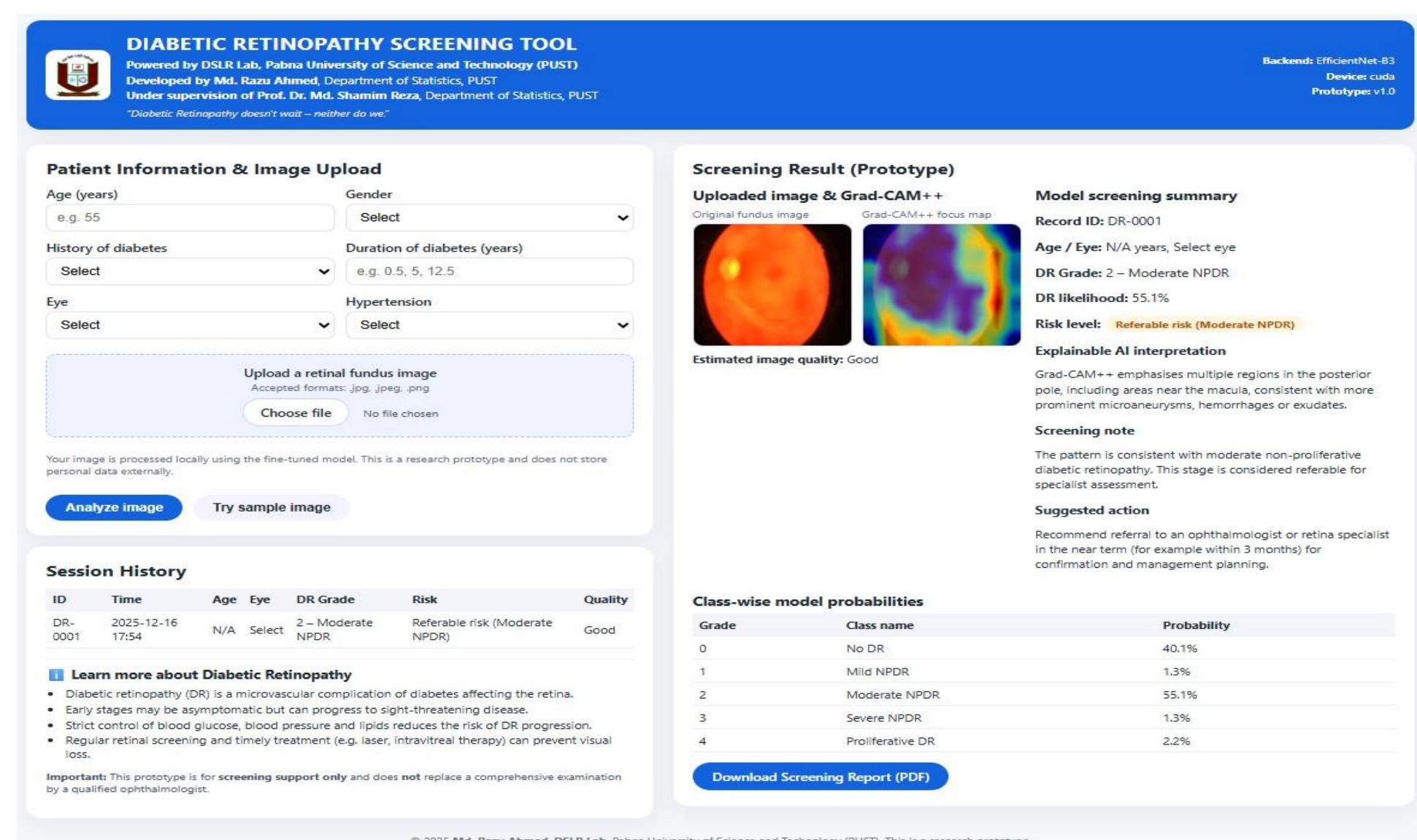


Figure 7: Prototype DR screening interface.

CONCLUSION

- ❖ A blended APTOS 2019 + IDRiD dataset with standardized preprocessing supports five-class DR grading under class imbalance.
- ❖ The best model, fine-tuned EfficientNet-B3 + SVM, achieves 72.7% accuracy, 58.4% macro F1, AUC 0.90, and QWK 0.59 on the blended test set.
- ❖ **Future work:** extend to larger local datasets and develop a mobile screening app based on this pipeline for real-world DR detection in low-resource settings.

Acknowledgment: This work was carried out at the Deep Statistical Learning & Research (DSLRL) Lab, Department of Statistics, Pabna University of Science and Technology (PUST) under the guidance of Dr. Md. Shamim Reza, and we thank the lab members for their support.

INTERNATIONAL CONFERENCE ON APPLIED STATISTICS AND DATA SCIENCE 2025 (ICASDS-2025)
DATA-DRIVEN SOLUTIONS: UNLOCKING NEW POSSIBILITIES

Scan for references and other information

